

Software Design Specification

IntegraCAD Open

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Client	[Client]
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Revision history

Rev	Date	Description	Author	Approved
0	2026-10-06	First issue	Llerd Grant	

Approval

Role	Name	Signature	Date
Author	Llerd Grant		
Reviewer (engineering)			
Approver ([Your company])			
Approver ([Client])			

1. Program organisation

POU	Type	Language	Called from	Interval	Purpose
Main	Program	LD	Cyclic task	10 ms	Coordination and calls
Safety	Program	LD	Safety task	10 ms	Safety logic only
Sequence	Function block	SFC	Main		Step sequencing
Alarms	Function block	ST	Main		Alarm evaluation
Analog	Function block	ST	Main	100 ms	Scaling and filtering
Comms	Function block	ST	Main	100 ms	External interfaces

Task allocation. [State why each POU is where it is. Anything that must react to a fast event goes in the fast task; anything that does not, does not, because scan time is a budget.]

2. Naming conventions

The convention matters less than the consistency. Pick one, write it here, and hold to it.

Object	Pattern	Example	Notes
Digital input	XX_nnn_FUNC	XV_101_ZSO	Device, number, function
Digital output	XX_nnn_CMD	MTR_101_RUN	
Analog input	XT_nnn	FT_101	ISA-5.1 letters
Internal flag	b prefix	bSequenceRunning	
Timer	t prefix	tValveTimeout	
Counter	c prefix	cCyclesComplete	
Constant	Upper snake	MAX_TEMP	
Function block instance	fb prefix	fbFeedPump	
Step number	STEP_	STEP_FILL	Named, never a bare integer

Forbidden. Single letter names, names that differ only by case, names that encode the address (I0_0), and any name that requires the comment to be readable.

3. Standard function blocks

The library used on this project. Each block is tested once and reused, rather than the logic being retyped per device.

Block	Purpose	Inputs	Outputs	Notes
FB_Motor	Motor start, stop, fault, run hours	Cmd, Fbk, Interlock, Reset	Run, Fault, Hours	Feedback timeout configurable
FB_Valve	Two position valve with limits	Cmd, ZSO, ZSC, Interlock	Sol, Fault, Position	Travel timeout per instance
FB_AnalogIn	Scale, filter, range check, alarm	Raw, RangeLo, RangeHi	Value, Fault, AlmL, AlmH	Detects out of range both ends
FB_Alarm	Alarm with delay, deadband, ack	Trigger, Ack	Active, Unack, Latched	
FB_Sequence	Step engine with timeout	Enable, StepDone	Step, Timeout	

Rule. A block is either general or it is not used. A "general" block with an `if device = 3` inside it is two blocks pretending to be one.

4. Memory and addressing

Area	Range	Use	Retentive
Inputs		Physical inputs	No
Outputs		Physical outputs	No
Markers		Internal flags	Partly
Retentive		Counters, run hours, recipe	Yes
Data blocks		Structured data	Per block

Retentive data. [List exactly what survives a power cycle and why. Anything retentive needs a defined behaviour on first run after a memory clear.]

5. State machine pattern

Every sequence uses the same shape, so that any engineer reading any sequence recognises it.

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STEP:

entry action, executed once

running action, executed every scan while in step

transition condition -> next step

timeout -> alarm and hold

...

Rules.

- A step is only left through a defined transition or a timeout. There is no other exit.
- Every step has a timeout. A step without one can hang forever, and the operator's report will be "it just stopped".
- Step numbers are named constants, never bare integers.
- The current step number is exposed to the HMI. Diagnosing a stuck sequence without it means going online with the programming software.

6. Coding rules

#	Rule	Reason
1	One coil per tag, in one place	Two coils on one tag means the last rung wins and the first is invisible
2	No duplicated logic, use a block	Two copies drift apart
3	No jumps backwards	Unreadable and can create loops
4	Every rung commented with intent, not restatement	"Start latches" not "XIC Start OTE Run"
5	No online edits without a change record	Undocumented online edits are how a program stops matching its printout
6	Safety logic in the safety task only	Mixing them defeats the safety certification
7	No forces left in the delivered program	A force is a temporary tool, not a fix
8	Timers use named presets, not literals	A literal 5000 explains nothing
9	Analog values checked for range before use	An out of range value used in a calculation propagates silently
10	First scan initialisation is explicit	Relying on default zero is relying on the vendor

7. Version control and change record

Version	Date	Author	Change	Reason	Tested	Approved
1.0	2026-10-06	Llerd Grant	First issue			

Backup. Source and compiled program archived at [location] after every change. Checksum recorded in the handover pack.